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Identification and selection of safety buffers in manufacturing companies

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ABSTRACT

The view on safety buffers is fragmented in the current literature; some researchers argue that a safety buffer is only waste, while others see them as prerequisites to absorb variations and secure a competitive delivery capability. This study conceptualises various safety buffer types in terms of materials, capacity and lead time to mitigate the negative effects of short-term stochastic variations in supply and demand. The identified safety buffers are categorised based on a material flow perspective as inbound, process and outbound buffers. In total, seven safety buffer sub-types are identified and investigated in terms of their utilization in four manufacturing companies. The experiences from eleven respondents highlight the utilization purposes in their selection of safety buffers. The empirical investigation also indicates several concerns through four propositions that highlight the significance of decision support, providing a more holistic perspective on different types and sub-types of safety buffers and their application in practice. Finally, a conceptual framework is proposed to facilitate the selection of safety buffers in practice.

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Safety buffer; buffer selection; stochastic variations; competitive delivery capability; buffer trade-offs

1. Introduction

The management of safety buffers is a challenge faced by most people involved in materials and production management. Among other things, it concerns decisions about the type of buffer, where to keep buffers, the size of each buffer, who is responsible for the buffers and the combinatorial effect of different buffers employed. A common denominator of all buffers is that they are used to mitigate disturbing variations. In many cases, a stable and completely predictable context for value-adding flows is assumed but this is not always a realistic assumption (Koh and Saad 2003; Luz et al. 2022; Rappold and Yoho 2008), in particular for complex environments (Fernández Campos, Trucco, and Huaccho Huatuco 2019). Variations represent deviations from expected patterns and can be external or internal in relation to a company (Hopp and Spearman 2008; Jafari et al. 2022; Luz et al. 2022; Ma et al. 2022). External variations may be present in terms of, for example, uncertain external demand or uncertain external supply lead times (Ben-Ammar, Bettayeb, and Dolgui 2020; Koh, Saad, and Jones 2002), while internal variations can be related to, for example, temporary capacity losses (Hopp and Spearman 2004; Goh et al. 2020; Ma et al. 2022). In addition, variations can have more long-term patterns that are of systematic character, such as seasonality, or be more short-term and superimposed on the systematic pattern (Goh et al. 2020; Wu and Holguin Veras 2022). The latter is called short-term stochastic variations. These short-term stochastic variations, which represent stochastic deviations from a systematic baseline, are considered in this study and for short here referred to as *stochastic variations*.

In addition to the classification of the variations as such, the causes of variations are important and can be classified as random or assignable. The impact of the former can be handled by buffers and the latter refers to causes that can be affected to reduce variations and thereby also reduce the need for buffers to handle all remnant variations (Hopp and Spearman 2021). Despite ambitious work with reducing the assignable variations, it is usually not possible to eliminate all these variations (Boute et al. 2008), and all remnant variations are considered as stochastic variations and the complete set of stochastic variations must be managed to maintain control (Koh, Saad, and Jones 2002; Romero-Silva, Santos, and Hurtado-Hernández 2022). These stochastic variations usually only result in minor disruptions and have not received much attention in the literature even though the impact on performance can be substantial due to their frequency (Wu and Holguin Veras 2022). Due to the significance of the impact on performance, the management of buffers is key and as Hopp and Spearman (2021) argue the buffers can be a combination in terms of materials, capacity, and lead time.

An addition of resources in the form of materials, capacity, and/or lead time to cope with stochastic variations in supply and demand is here referred to as a *safety buffer*. The amount and types of resources in a company are prime economic factors and decisive in the manufacturing abilities and

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limitations (Tortorella et al. 2022; Van Mieghem 2003). The utilization of safety buffers might improve performance, but to be of real value, the benefit of safety buffers must exceed the costs incurred of for example capital tied-up in additional resources (Hedvall, Wikner, and Hilletofth 2017). Consequently, there is a need to manage safety buffers involving different strategic aspects such as selecting safety buffer types (materials, capacity or lead time), positioning of safety buffers in the value-adding flow, and dimensioning of safety buffers. This approach to safety buffers is here referred to as *safety buffer management*. Stochastic variations are to some extent present in all companies and both the number of and the size of the safety buffers tend to increase if they are not actively managed with continuous attention (Hopp and Spearman 2004). Safety buffers can be individually managed as in traditional inventory control or in a hierarchical fashion as in MRP or even using an adaptive approach (Wikner, Hedvall, and Mattsson 2019). All cases do however require that the safety buffers are identified. Hedvall (2018) provided preliminary findings of safety buffers with a capacity management perspective, although lacking a theoretical foundation and being inconclusive with respect to providing support in the selection of safety buffers in practice.

A comprehensive approach to safety buffer management in general, and to the selection of safety buffers in particular, is not well covered in the literature. As a result, there is a lack of both coherent terminology and an integrated perspective on different types of safety buffers. Many studies connected to safety buffers are based on the work by Whybark and Williams (1976) with different perspectives. There have been relatively limited contributions in this area in recent years, especially in relation to the development of theory application in practice (Gonçalves, Sameiro Carvalho, and Cortez 2020). Guide and Srivastava (2000) pinpointed that there is a lack of agreements and conflicting results for which safety buffers to use, with no clear resolution for the size of the safety buffers, and also general guidelines for safety buffer management are difficult to find. Since then, van Kampen, van Donk, and van der Zee (2010) investigated the impact of the variations on delivery performance when using safety stock in comparison to using safety lead time with a simulation study and found that they are preferable for different types of variation, but they observed that safety lead time is superior when facing variations in supply and demand simultaneously. Ben-Ammar and Dolgui (2018) investigated optimal order release dates to suppliers based on stochastic lead times but with known demand and infinite capacity, where the release date is optimised based on a backorder cost and inventory holding cost. Hence, a safety lead time is compared to the cost of backorders. Kasper, Land, and Thürer (2023) recently investigated control priorities for parallel machines, which is highly relevant in the context of variations and safety buffers, but the relation of these priorities to the selection and dimensioning of safety buffers is not obvious.

The limited coverage in the literature was evident in a research project where it was obvious that these studies do not provide a rationale for *safety buffer selection* when companies choose one type or the other of safety buffers and how safety buffer management should be handled in practice. The approaches most used in the literature related to safety buffers are mathematical modelling, simulation modelling, and scheduling heuristics (Koh, Saad, and Jones 2002). The qualitative perspective through case research has not received substantial interest in the literature (van Kampen, van Donk, and van der Zee 2010). A notable exception is the research by Land et al. (2021) that applied a qualitative perspective with a focus on identifying the causes for buffers and safety buffers by a design science approach for diagnosing flow problems. This study extends the research by Land et al. (2021) and focuses on the safety buffers to choose from and their purposes in the safety buffer selection given existing causes for stochastic variations. This is sought to address the challenge of transferring theoretical knowledge to actual practical settings, as emphasized by Romero-Silva, Santos, and Hurtado-Hernández (2022). In summary, the purpose of the study presented here is to investigate a qualitative perspective of safety buffer selection to highlight the practical perspective and contribute to this area of research.

Our study therefore takes a stance in safety buffer management in practice with the objective to provide support for safety buffer selection by developing a conceptual framework of safety buffers and investigating their use in some manufacturing companies. The study consists of three main parts: first a conceptual framework of safety buffers and their purposes is developed, and second an empirical investigation in four manufacturing companies is performed. Third, a conceptual framework to support the selection of safety buffers in practice is proposed based on the literature and empirical investigation. An important aspect in the empirical investigation has been to investigate how safety buffers are used in different situations, where safety buffers can have different purposes from a material flow perspective. In this study, safety buffers with distinct purposes from a material flow perspective are referred to as different safety buffer sub-types. Three main research questions (RQ) have guided the empirical discussions: RQ (i) What are the main safety buffer sub-types utilized in manufacturing companies?; RQ (ii) What is the purpose of utilizing these safety buffer sub-types in manufacturing companies?; and RQ (iii) What are the main concerns in the selection of safety buffers in manufacturing companies?

Next, a description of the research methodology employed for this study is presented. This is followed by outlining the safety buffer terminology from a material flow perspective. Thereafter, the literature on safety buffers to form the conceptual framework employed in the empirical investigations is presented. Finally, the empirical findings are presented, a conceptual framework based on the findings is proposed and some concluding remarks are made.

2. Research methodology

For the conceptual framework, a method of analytical and logical reasoning is employed by using input from research

literature related to safety buffer management. The literature is combined with the researchers' experiences from industry to develop a framework of safety buffer types and sub-types for stochastic variations in supply and demand. Logical reasoning can provide new insights into a field, with opportunities to develop new concepts and conceptualize relationships (Meredith 1993). In addition, it provides opportunities to explain, provide an understanding, or express conceptual perspectives on theory (Wacker 1998).

2.1. Literature review

The relationships of interest in this study, which are combined in this paper, are between the type of variation in supply and demand (stochastic) and types of safety buffers (material, capacity, or lead time). Different sub-types of safety buffers in the form of material, capacity, and lead time are identified based on existing literature and synthesized into a framework. The purpose of the literature review was to acquire insight into the subject of interest to subtract key components for the conceptual framework. The Scopus database has been used to retrieve research articles. To exemplify, search words used to retrieve records for capacity safety buffers (i.e. safety buffers in the form of capacity) were, among others: 'safety capacity', 'slack capacity', 'capacity buffer' and 'protective capacity'. Snowball sampling has also been used, where relevant references from the original sample have been included.

2.2. Empirical investigation

The empirical investigation originates from empirical observations in a research project. These empirical observations constitute a baseline for analysis and have primarily been conducted through workshops and interviews with respondents from the participating companies in the research project. The focus of this part of the study was to investigate the selection of safety buffers in manufacturing companies to make a first evaluation of the conceptual framework. The unit of analysis is limited to the selection of safety buffers and the investigation is based on the experiences of 11 respondents at four manufacturing companies.

A qualitative direction is considered an expedient choice for this study, as it enables the possibility to explore and generate in-depth knowledge about the phenomenon of interest. According to Voss et al. (2002), interviews and documents are the most common sources of information in qualitative research. The data collection in this study included both sources but with a primary source in the form of two interactive workshops that included all four companies. The companies received an assignment with questions to discuss internally before the workshops, see Appendix A for a summary of the assignments. Note that the workshops had a broader scope than the study presented in this paper and the questions presented in Appendix A are therefore the questions of relevance for the scope of this paper. The assignments for the workshops explicitly included the first two research questions that have guided the empirical investigations, while the third research question has been implicitly included by the questions asked during discussions in the workshops and interviews. The discussions often included a set of why-questions (e.g. why do you have this safety buffer sub-type? Why do you think this safety buffer sub-type is used in another way than intended? We can see that there are many safety buffer sub-types within the company, reflect on why you think this is the case?).

During the first workshop (workshop 1), one of the researchers provided a lecture on safety buffer types and sub-types in general and how variations can be reduced as a complement of using safety buffers. The conceptual framework of safety buffer sub-types was thereafter included and described briefly in the assignment for the second workshop (workshop 2). Both the lecture and conceptual framework were sought to develop a common terminology used within the research project to reduce the risk of misunderstandings and misinterpretations. The companies presented their answers and work connected to the questions in the assignments during the workshops, where the compounded presentations constitute important documentation.

Before the interviews, workshop recordings and the documents were scrutinized to summarise the remaining questions into an interview guide. Depending on the level of detail in the presentations and what was mentioned or discussed during the workshops, the remaining questions varied from company to company. Hence, the interview guide was unique for each company but the purpose of the interviews was the same; to get a deeper understanding of the selection of safety buffers and resolve obscurity from previous presentations and discussions.

The case companies' representatives were interviewed through semi-structured, open-ended interviews to identify safety buffer types and sub-types and the work with these in practice. Enhanced reliability (as described by Lincoln and Guba 1985) is sought through audio-recorded interviews, where relevant information has been extracted and transcribed. Companies from different industries have been selected based on the overall intention of the research project, which enables a broadened understanding of the selection of safety buffers in this study. The main criteria for inclusion in this study were: (1) manufacturing company, (2) working actively with safety buffers and willing to study their work in a research setting, and (3) having both standardized and customized products. These criteria were used to ensure that safety buffers connected to manufacturing processes could be investigated (1), that the involved respondents would have insights and the possibility to discuss with each other their selection of safety buffers (2), and that standardized products are often connected to a focus on inventory and customized products to a higher degree of focus on capacity and lead times, therefore providing both opportunities for investigating different types of safety buffers at each company and involving companies with experience from several types of safety buffers (3).

Table 1. Empirical data collection.

Company	Product	Employeesa	Turnover th. EUR	Company size	Respondent	Position	Method	Duration
Company A	Gas turbines	2605	1,003,796	Very large	R1	Manager Sales and Operations	Workshop 1	10 h
							Workshop 2	10 h
							Semi-structured interview	2 h 10 min
					R2	Procurement Director	Workshop 2	10 h
					R3	Leader Logistics Development	Workshop 1	10 h
							Workshop 2	10 h
					R4	Capacity Planner	Workshop 1	10 h
							Workshop 2	10 h
Company B	Luminaries	581	150,444	Very large	R5	Team Leader Planning	Workshop 1	10 h
							Workshop 2	10 h
							Semi-structured group interview	2 h 30 min
					R6	Production Planner	Workshop 2	10 h
							Semi-structured group interview	2 h 30 min
					R7	Logistics Developer	Workshop 1	10 h
Company C	Heat exchangers	136	52,999	Large	R8	Supply Chain Manager	Workshop 1	10 h
							Workshop 2	10 h
							Semi-structured interview	2h 45 min
					R9	Production Planner	Workshop 1	10 h
Company D	Aerospace components	2131	1,178,977	Very large	R10	Senior Specialist Logistics	Workshop 1	10 h
							Workshop 2	10 h
							Semi-structured group interview	3h 10 min
					R11	Process Developer	Workshop 1	10 h
							Workshop 2	10 h
							Semi-structured group interview	3h 10 min

[a]The Amadeus database was used to retrieve records concerning the number of employees, turnover and company size based on data from 2019 and 2020

Table 1 summarises information about the companies and data collection in the face-to-face meetings, where company names are changed to Company A, Company B, and so forth for anonymity. *R* is hereafter used to designate the respondent, with the respondent number as indicated in the table. The duration refers to the duration of the complete workshop. Additional questions arising after the interviews were handled by further contact with the respondents. The interviews were performed in the native language of both the researchers and the respondents to avoid language barriers, encourage discussion, and reduce the risk of misunderstandings.

An interview review framework was established by extracting relevant information from workshops, interviews, and additional documents. This interview review framework consists of the identified safety buffer subtypes and the main descriptions and purposes for each sub-type based on explanations from respondents, which, together with quotes from respondents, exemplify connections. The data analysis to retrieve the interview review framework mainly included to examine, compare, contrast and categorize company descriptions in relation to the conceptual framework. This means that main data codes were established a priori and augmented the analysis, as recommended by Miles, Huberman, and Saldaña (2014). Empirical data from the companies were treated separately before comparison to each other, also referred to as in-case and cross-case analysis as described by Eriksson and Kovalainen (2008). An important part was to control that company descriptions in presentations and from recordings corresponded to the definitions in the conceptual framework, to ensure correct categorisation of data. The purposes and main concerns described by the companies were categorised based on the safety buffer types and sub-types to facilitate comparisons across the companies.

Eisenhardt (1989) emphasized the importance of a synergetic view of evidence to increase the evidence in the findings. The synergetic view of evidence is sought to be strengthened here by multiple data collection methods that enable a deeper understanding and triangulation of findings. Williamson (2002) presented different types of triangulation, where method triangulation and source triangulation have been used in this study. Method triangulation has been used in this study by different data collection methods (workshops, assignment presentation slides, and interviews) and source triangulation (as described by Williamson 2002) by multiple respondents within each company. In addition, multiple researchers have been involved, where one of the researchers has summarised findings that have been reviewed by the other researchers to check consistency in the findings and enhance the likelihood of capitalising novel insights. There are often complementary insights, making this iteration between researchers important. Audio-recorded workshops and interviews provide an accurate rendition of what has been said, which, together with documentation of the research, are used to enhance the research reliability, as described by Lincoln and Guba (1985).

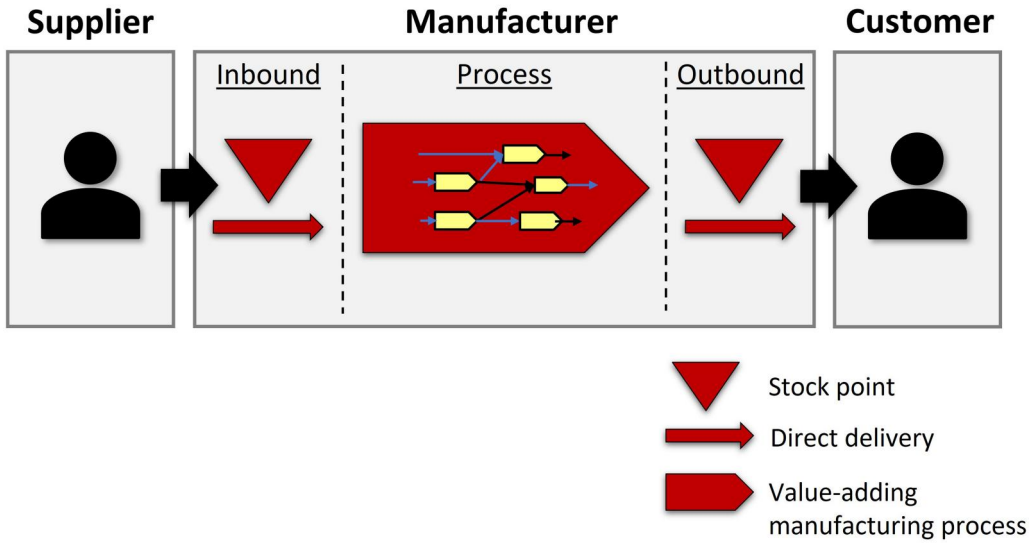


Figure 1. Material flow perspective.

3. Safety buffer terminology

Buffer, as a general concept, is here defined as something that serves as a protective barrier. For manufacturing companies, such a barrier has the purpose of protecting the delivery capability of the business (Oehmen et al. 2009; Romero-Silva, Santos, and Hurtado-Hernández 2022), particularly the support of resources and their contribution (Gólcher-Barguil, Nadeem, and Garza-Reyes 2019). Hopp and Spearman (2004) argued that capacity, material, and lead time are the three fundamental types of safety buffers for stochastic variations, while Caridi and Cigolini (2002) referred to three main slack resources for variations. In both cases, the buffers concern materials (inventory); capacity; and lead time (time), which are here referred to as safety buffer types when concerning buffers for stochastic variations. All three safety buffer types are considered in this study and, adapted from the definitions by Hopp and Spearman (2004), are defined as (i) material safety buffers representing an additional inventory of input materials, semi-finished items, or finished products as a hedge against stochastic variations in demand, delivery from suppliers and/or in manufacturing, (ii) capacity safety buffers representing additional capacity (sometimes referred to as surplus capacity) or flexibility (e.g. multiskilled workers and machines) as protection against stochastic variations in demand and/or manufacturing capacity, and (iii) lead time safety buffers representing an increase in lead time from the absolute minimum to an extra amount of lead time to hedge against stochastic variations in delivery lead times from suppliers and in manufacturing processes. In this context, the emphasis is on extending the lead time at an item level. Hence, as described in the introduction, a safety buffer is here interpreted as an addition of resources in the form of material, capacity, and/or lead time to cope with stochastic variations in supply and/or demand.

From a material flow perspective, the requirements of available materials and the capacity used to fulfil demand can be illustrated as in Figure 1. Note that the time aspect is implicitly covered by the timing of customer demand and the lead times related to external supply and the internal value-adding manufacturing processes. The material flow perspective represents any kind of operation in a supply chain, converting input from suppliers into output to customers (Stefansson, Jensson, and Shah 2009; Rasouli et al. 2019).

Figure 1 encompasses three significant parts for the manufacturer: the inbound, process, and outbound of the company. From the manufacturer's perspective, inbound represents an interface between the supplier (external or internal manufacturing process upstream the material flow) and the internal value-adding manufacturing processes (Rasouli et al. 2019). It covers the replenishment of raw materials and components to inventory on the upstream side of manufacturing or directly to manufacturing orders (Stefansson, Jensson, and Shah 2009). The process in Figure 1 represents the manufacturer's internal value-adding manufacturing processes with resources in terms of personnel and equipment for supply. Outbound, finally, represents an interface between the value-adding manufacturing processes and delivery to the customer (Rasouli et al. 2019). It covers the inventory of finished products and delivery on the downstream side of the manufacturer (Stefansson, Jensson, and Shah 2009). From a material flow perspective, the safety buffer types have distinct purposes (van Kampen, van Donk, and van der Zee 2010), leading to different safety buffer subtypes which are further discussed in the conceptual framework. Henceforth, safety buffers as terminology is used for discussions regarding safety buffers in general, covering both safety buffer types and sub-types, while safety buffer types or safety buffer sub-types are used when referring specifically to one or the other.

4. Frame of reference

Stochastic variations are superimposed on the systematic variations of demand per period. The magnitude of

stochastic variations is typically estimated as ranges in percentage around the average regular demand or calculated as a measure of dispersions such as the standard deviation. Kechaou, Addouche, and Zolghadri (2022) listed several specific company losses, such as the non-scheduled time when a machine is out of service, standby time for lack of components, reduced speed, and minor stoppages. All these are related to stochastic variations in supply or demand and it is important to be aware of the existence of stochastic variations in order to take appropriate measures (Stefansson, Jensson, and Shah 2009) and find appropriate safety buffers to absorb them (van Kampen, van Donk, and van der Zee 2010). Guide and Srivastava (2000) proposed dampening methods to reduce the effects of variations, consisting of order management and the use of slack resources. The former alternative is beyond the scope of this study. Thus, the focus here is on managing variations by slack resources (i.e. safety buffers) for stochastic variations in supply and demand. The three safety buffer types (material, capacity and lead time) with their underlying differentiations of safety buffer sub-types are presented and characterised in the following sections to form the conceptual framework in section 5.

4.1. Material safety buffers

Traditionally, efforts have focused on reducing production costs by economies of scale, leading to holding inventories to maximize resource utilization (Fry, Steele, and Saladin 1994; Gólcher-Barguil, Nadeem, and Garza-Reyes 2019). According to Fry, Steele, and Saladin (1994), the primary role of inventory is to protect the manufacturing system from unexpected fluctuations. A safety stock is an added quantity of materials to hedge against the risk of stockout due to stochastic variations in supply or demand (Angkiriwang, Pujawan, and Santosa 2014). The literature on the dimensioning of safety stocks is vast, while the contributions covering the positioning and managing of these safety buffers is less extensive. However, according to the theory of constraints, time buffers should be carefully positioned for example before and after the bottleneck resource, i.e. management with a focus on resources that constitute constraints on the material flow (Telles et al. 2020; Thürer et al. 2022). Formally, however, the time buffers of TOC are more related to lead time safety buffers and further discussed below. In general, there are two main sub-types of material safety buffers for stochastic variations. The first sub-type consists of raw materials or components from suppliers or semi-finished materials from internal manufacturing processes, which protect the manufacturing system when higher consumption than expected arises due to stochastic demand variations (Kasper, Land, and Thürer 2023). From a material flow perspective, this safety buffer sub-type is related to the inbound side and is hence referred to as an inbound safety stock buffer. The second sub-type consists of finished goods and protects the manufacturing system and deliveries to customers from unexpected customer demand and unexpected loss of manufacturing capacity (Fry, Steele, and Saladin 1994). This safety buffer sub-type is instead on the outbound side and is referred to as the outbound safety stock buffer.

An important notion for material safety buffers is that they can be an expensive proposition when holding costs are high (Hu, Duenyas, and Kapuscinski 2003). However, there are several advantages of holding safety stocks when managed properly, including protecting against potential shortages due to stochastic variations in demand and/or supply (Kanet 2014). This holds especially for avoiding the amplification of variations due to over-reactions to short-term variations that are caused by adjustments of parameters based on inventory replenishments rules (Boute et al. 2008).

4.2. Capacity safety buffers

In relation to materials, capacity represents the potential of more flexible safety buffers (Jafari et al. 2022), in that it can be applied for a wider set of demand patterns and possible to use for manufacturing and delivery of several different materials compared to a material safety buffer (van Kampen, van Donk, and van der Zee 2010). The downside is that available materials can be used immediately for delivery, whereas available capacity requires a certain lead time before delivery is ready and there is a risk of resource loss when not used (Fry, Steele, and Saladin 1994). Bradley and Glynn (2002) argued that overly restricting the capacity level causes the total cost to increase towards infinity, and capacity may not always be so expensive that high capacity utilization must be warranted (Romero-Silva, Santos, and Hurtado-Hernández 2022). It was further described that the utilization level affects the business responsiveness. Marucheck and McClelland (1992) investigated the impact of utilization rates on different performance measures, where all performance measures tend to decrease as capacity utilization increases (except for low utilization between 50-80%). The measures deteriorate exponentially for a utilization rate over 80%, with a significant decline for extremely high utilization rates between 95 and 98% (Marucheck and McClelland 1992). To avoid these negative effects on performance, the theory says that it is always necessary to establish capacity safety buffers to hedge stochastic variations in demand and available capacity (Caridi and Cigolini 2002). In this context, Wu and Holguin Veras (2022) found labour shortage as the most important factor regarding disruptions (i.e. variations) with the most impact on daily output and disruption duration, highlighting the need for capacity safety buffers.

In the review of the literature, nomenclature varies in terms of what additional capacity is called when used as a hedge against stochastic variations. In this study, this safety buffer sub-type is referred to as safety capacity. Additional designations used in the literature considered to cover the same topic include protective capacity as described by Atwater and Chakravorty (2009) and Goldratt (1990), capacity buffer (Bradley and Glynn 2002), capacity cushion (Blackstone 2013), idle capacity (Goodale, Verma, and Pullman 2009), excess capacity (Atwater and Chakravorty [2009]; Wu, Erkoc, and Karabuk [2005]; Fry, Steele, and Saladin [1994]) and slack capacity (Romero-Silva, Santos, and

Hurtado-Hernández 2022). Safety capacity is a hedge against uncertainty in supply and demand to catch up with the flow of materials or to absorb peak demand (Fry, Steele, and Saladin 1994; Kasper, Land, and Thürer 2023). Fry, Steele, and Saladin (1994) argued that safety capacity is not merely a cost; for personnel, it has additional value if effectively used for continuous improvement, such as addressing problems with variations and reducing requirements of safety buffers. This means that when the safety buffer is not used to absorb variations, in particular the unused capacity of man hours can be used to reduce variations through continuous improvements and thereby reduce the need for safety buffers.

Capacity flexibility is essential to achieve a high service level and efficiency (Betts, Meadows, and Walley 2000; Pagell et al. 2000), where the manufacturing system can handle a larger range of volume and/or variety at an affordable cost (Razak, Hendry, and Stevenson 2023). According to Bish, Muriel, and Biller (2005), flexibility in machinery can be used to hedge against variations in customer orders in the short term. Fry, Steele, and Saladin (1994) emphasized the requirements of flexibility and short delivery lead times for companies that compete on fast delivery of existing products or the introduction of new products. Productive resources can be permanent or contingent. Permanent represents the internal resources of a company, the workforce level on a steady payroll and the machinery owned (Tan and Alp 2009). The total capacity level can be increased by acquiring contingent resources, representing flexibility (Rappold and Yoho 2008), here referred to as a capacity flexibility buffer, such as authorizing overtime, multiskilled personnel that can relocate on a daily basis depending on the needs, or machines that can handle several products (Rappold and Yoho 2008). Here, a capacity flexibility buffer is regarded as a sub-type of capacity safety buffers to achieve capacity flexibility by different contingent resources.

4.3. Lead time safety buffers

According to Pittman and Atwater (2016), lead time is the time span required to perform one or several operations. A short delivery lead time (i.e. the lead time from the received customer order to delivery) is commonly seen as a competitive advantage (De Treville *et al.* 2014), as short delivery lead time represents an element of responsiveness for quick response to stochastic variations in demand (Iyer and Bergen 1997; Jafari et al. 2022). However, unpredictable supply or demand increases the value of additional lead time to absorb stochastic variations (Buzacott and Shanthikumar 1994; Krishnamurthy, Suri, and Vernon 2004). Ben-Ammar, Bettayeb, and Dolgui (2020) found that when variations in lead time cannot be reduced or precisely estimated, it is better to overestimate the lead time to prevent the total costs from becoming much higher than expected. In this study, an additional amount of lead time to absorb stochastic variations is referred to as a lead time safety buffer. The lead time safety buffers aim to ensure that input materials are available in time for manufacturing, that unexpected events in manufacturing can be managed in time before delivery (Kasper, Land, and Thürer 2023), to ensure that the output from manufacturing is available in time for delivery, or to smooth the capacity requirements (Angkiriwang, Pujawan, and Santosa 2014).

A lead time safety buffer can be used to absorb stochastic variations in delivery lead time from suppliers, processing times in manufacturing, indirect time, available capacity or due to stochastic variations in demand (van Kampen, van Donk, and van der Zee 2010). Lack of information and knowledge sharing with suppliers is connected to uncertainties in deliveries and actual delivery delays (Gallear et al. 2022), highlighting the need for lead time safety buffers.

When used in the procurement of materials, it implies ordering in advance of actual need to ensure that materials are available in time for the planned start of manufacturing orders, therefore ensuring that the capacity is not starved due to material shortages (Angkiriwang, Pujawan, and Santosa 2014; Zhai et al. 2017). This additional lead time is here referred to as an inbound safety lead time buffer. Lead time safety buffers are also applied when manufacturing orders are planned to start in advance of what is required in order to have a safety lead time before delivery to a customer or an inventory so any stochastic variation in demand or manufacturing processes do not cause delivery delays (Angkiriwang, Pujawan, and Santosa 2014; van Kampen, van Donk, and van der Zee 2010), referred to as an outbound safety lead time buffer. The downside with safety lead time is that it increases the total lead time, increase capital tied-up in work-in-process, and can create distorted priorities (Chu and Hayya 1988), with a risk that set dates for deadlines lose credibility as the actual requirements are hard to detect in the planning system.

A queue time is the amount of time spent waiting before manufacturing processes are performed (Lantz and Rosén 2017). Queue time is here referring to a stochastic variable, while queue lead time is a decision parameter representing an added amount of lead time to hedge stochastic variations. Vollmann et al. (2005) explained that queue lead time is an important addition incorporated into manufacturing processes, as it works as a hedge against unexpected events. A queue lead time buffer is here regarded as a provision of a wider time span to manage both the planned work and time in the queue. The actual queue time is influenced by the resource utilization rate and determines the queue lead time and, consequently, the delivery lead time (Lantz and Rosén 2017). Gunasekaran, Patel, and Tirtiroglu (2001) emphasised the importance of measuring and controlling capacity utilization, as it directly affects the queue time and responsiveness to customer demand. It was further pointed out that the measurement is a prerequisite to achieving improved responsiveness and reduced delivery lead times. Goldratt introduced a similar buffer concept, see e.g. Goldratt (1990) and (Goldratt 2009), referred to as a 'time buffer'. A time buffer is however an aggregate concept that represents the sum of the operation time for a sequence of operations either upstream, downstream or parallel to a constraining resource, and in addition to this some slack time. In this sense the

operations in a time buffer 'share' queue time. According to the theory of constraints, see e.g. Cox and Schleier (2010), such time buffers are positioned to protect the material flow through the system which then is controlled by the constraining resource. This approach is basically related to how to combine safety buffers which is an important extension of individual safety buffers, but not in focus here where different types of safety buffers are key.

4.4. Safety buffer selection and trade-offs

As early as 1988, Chu and Hayya (1988) argued that there is a lack of general guidelines for the selection of safety buffers even though there are several safety buffer types and sub-types to hedge against variations. However, rules of thumb are mentioned in some studies on how to apply different types of safety buffers in manufacturing systems. New (1975) showed that lead time safety buffers may be more beneficial for lower-level items, in the bill-of-materials, and material safety buffers for end-items. Whybark and Williams (1976) simulation study indicated that lead time safety buffers are preferred for variations in terms of timing, and material safety buffers are preferred for variations in quantity. A simulation study by Schmitt (1984) included a capacity safety buffer; the results indicated that the best safety buffer type depends on the source of variation and the costs associated with either choice. The use of capacity safety buffers alone as a strategy to achieve performance goals is questionable for high demand variance in the product mix (Marucheck and McClelland 1992). For this situation, it would probably be more suitable to include safety lead time in customer quote dates, add material safety buffers and/or work towards reducing variations in procurement, processing times or forecasting (Angkiriwang, Pujawan, and Santosa 2014; Marucheck and McClelland 1992). This illustrates that which safety buffers to consider depends on the surrounding circumstances and reasons for utilizing a certain type and sub-type of safety buffer. Lead time safety buffers may lead to high on-time delivery performance but also result in longer delivery lead times (Angkiriwang, Pujawan, and Santosa 2014), which may erode competitiveness if delivery speed is of high importance. The combination of selected safety buffers constitutes the company's ability to absorb stochastic variations, where these trade-offs need to be considered to avoid negative effects on competitiveness. See, e.g. Rajagopalan (1998) for a study on the trade-off between using excess capacity for increasing the number of changeovers to reduce inventory levels and having more inventory to postpone future requirements of capacity investments. Steele and Papke-Shields (1993) and Angkiriwang, Pujawan, and Santosa (2014) argued that a safety capacity buffer is a competitive weapon that can reduce lead times and the need for inventory. This is also in line with the theory of constraints where a balance between protective capacity in non-constraints and the length of the corresponding time buffer is key. South and Hixson (1988) also argued that it may significantly reduce the level of work in process inventory and is a prerequisite for flexibility and responsiveness. Atwater and Chakravorty (2009) emphasized that there is a need to better understand the requirements of safety capacity buffers as a protection, especially with an emphasis on minimizing waste in the operating environments.

To some extent, the different safety buffer types are interchangeable with each other, which implies that the selection of safety buffer type and sub-type is of strategic importance. This is especially the case as the selection of safety buffers affect the delivery capability (Boute et al. 2008). There are some general trade-offs between different safety buffer types (see, e.g. Vollmann et al. (2005), as illustrated in Figure 2. The general conclusion from Figure 2 is that different safety buffer types are mutually dependent in the sense that the utilization of one safety buffer type decreases the need for another. The red line in each graph illustrates that certain customer service can be provided by a combination of two safety buffer types.

The problem of finding an appropriate combination in terms of safety buffer types, positions, and sizes remains. According to Hurley (1996), this is a 'complex combinatorial problem', which has led to a tendency to simplify matters in the literature by e.g. decoupling using buffers and safety buffers between work stations in the studied systems, assuming material availability and that all necessary information such as processing times is known (see e.g. Thürer et al. [2022]).

5. A Conceptual framework of safety buffers for stochastic variations

A framework of safety buffers is proposed, based on the frame of reference above. The three different safety buffer types outlined in the frame of reference are all important for different situations. When reviewing nuances of material, capacity, and lead time safety buffers, seven different safety buffer sub-types have been identified. In Table 2, the framework is illustrated with safety buffer types, safety buffer sub-types, purposes, and descriptions. The safety buffers are

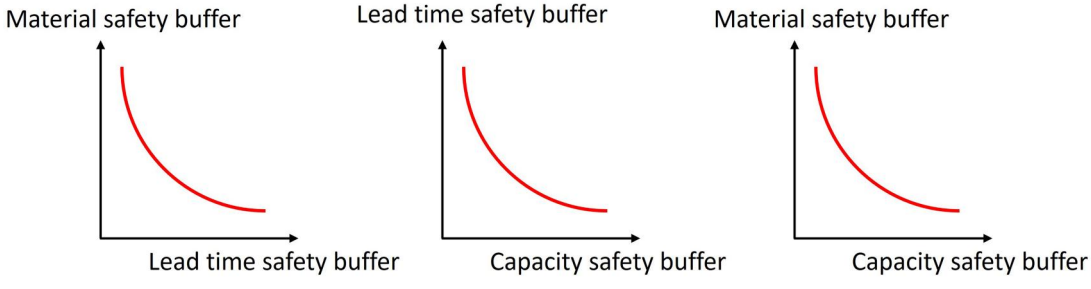


Figure 2. Safety buffer trade-offs.

Table 2. Safety buffers for stochastic variations in supply and demand.

Category	#	Safety buffer type	Safety buffer sub-type	Purpose	Description
Inbound buffers	B1	Material	Inbound safety stock buffer	Protects process capacity against material shortages	An inbound safety stock buffer is an extra quantity of materials placed in front of value-adding manufacturing processes to protect from material shortages, providing a protection of the planned order start and available resource capacity to utilise fully.
	B2	Lead time	Inbound safety lead time buffer	Protects process capacity against material shortages	An inbound safety lead time buffer is an added amount of time in front of value-adding manufacturing processes to protect the planned order start and available resource capacity to utilise fully and in front of net material requirement due dates to protect inventory on hand from shortages.
Process buffers	B3	Capacity	Safety capacity buffer	Allows more stable, short and reliable delivery lead times	A safety capacity buffer is an amount of capacity added to the average required capacity to allow stable, short and reliable delivery lead times.
	B4	Capacity	Capacity flexibility buffer	Short-term opportunities to increase capacity when needed	A capacity flexibility buffer consists of short-term opportunities to increase capacity beyond the safety capacity buffer, e.g. utilise overtime, multi-skilled employees.
	B5	Lead time	Queue lead time buffer	Protects delivery lead time	A queue lead time buffer is the amount of time a job is planned to wait at a work centre before setup or work is performed on the job, smoothing the capacity requirements as there is a wider time span to manage the work.
Outbound buffers	B6	Material	Outbound safety stock buffer	Absorbs variations in demand and compensates insufficient process capacity	An outbound safety stock buffer absorbs variations in demand and compensates for insufficient resource/work centre capacity to allow high service levels.
	B7	Lead time	Outbound safety lead time buffer	Absorbs variations in capacity and compensates insufficient process capacity	An outbound safety lead time buffer absorbs variations in capacity and compensates for insufficient resource/work centre capacity to allow more stable and reliable delivery lead times.

categorized from a material flow perspective as inbound, process or outbound buffers and further described below. B is hereafter used to designate the safety buffer sub-type together with a number to be used as a reference for the specific safety buffer sub-type.

Inbound buffers represent safety buffers that protect the manufacturing processes from starvation of materials, due to unexpectedly high demand or unexpected disruptions in delivery of input materials, that inevitably might cause capacity losses, i.e. unused capacity. The protection of the manufacturing system was emphasized by Fry, Steele, and Saladin (1994) and Angkiriwang, Pujawan, and Santosa (2014), which further pinpointed the importance of a material safety buffer for materials delivered from suppliers. This material safety buffer sub-type, when an extra amount of materials protects the resource/work-centre against material shortages, is here referred to as an inbound safety stock buffer (B1). Note that for consistency the buffer types use a similar naming structure, i.e. material safety buffer, capacity safety buffer, and lead-time safety buffer. In some cases, the naming of the safety buffer sub-types is similar to the safety buffer types but the buffer type names all end with 'safety buffer' which is not the case for any sub-type. An alternative to having additional materials as a protection is to order input materials in advance of requirements, i.e. using an inbound safety lead time buffer (B2). This safety buffer sub-type can also be used when suppliers deliver specific quantities to be used directly by manufacturing orders. Inbound safety lead time buffers imply that an additional amount of time is added to the planned lead time to secure the delivery of materials before the planned requirements (Angkiriwang, Pujawan, and Santosa 2014).

The process buffer category in Table 2 represents safety buffers that protect the customer of a manufacturing process, or the receiving inventory, from late deliveries due to stochastic variations and disruptions. Fry, Steele, and Saladin (1994) argued that an important addition of resources in terms of capacity can absorb unexpected events or peak demand. Atwater and Chakravorty (2009) and Huang, Song, and Tong (2016) added that this type of additional capacity is of strategic importance, especially for protecting the delivery capability (Thürer et al. 2017). Thus, additional available capacity can allow more stable, short, and reliable delivery lead times and is here referred to as a safety capacity buffer (B3).

One other sub-type of capacity safety buffers is closely connected to the safety capacity buffer, with the difference that the additional amount of capacity is called for when needed by flexibility in resource capacity (Sarker et al. 2022). This sub-type of capacity safety buffer in the manufacturing processes is related to what Van Mieghem (2003) call operational hedging, in other words, utilizing various types of flexibility to absorb stochastic variations in supply and demand (Huang, Song, and Tong 2016;

Rappold and Yoho 2008; Razak, Hendry, and Stevenson 2023). To temporarily increase the capacity level by contingent resources (as described by Tan and Alp 2009) represent the short-term opportunities of a capacity flexibility buffer considered in this study, see B4 in Table 2. A capacity flexibility buffer is related to an agility strategy but is here seen as reserve capacity that in general, is permanently available to be utilized temporarily when needed. The last process buffer considered in this study is referred to as a queue lead time buffer (B5), which also protects the delivery lead time from unavoidable queue times in front of work centres by a wider time span rather than by having additional capacity. Queue lead time buffers are commonly applied in manufacturing processes (Vollmann et al., 2005), smoothing capacity requirements and making the delivery lead time longer, but also more predictable (van Kampen, van Donk, and van der Zee 2010).

Outbound buffers include the same types of safety buffers as the inbound buffers, with the difference that the safety buffers substitute the need for additional capacity safety buffers to absorb stochastic variations in supply and demand, protecting the delivery capability (Angkiriwang, Pujawan, and Santosa 2014). When the protection of the delivery capability is in the form of additional material, it is here seen as an outbound safety stock buffer (B6), while it is called an outbound safety lead time buffer (B7) for an additional amount of lead time before delivery.

6. Empirical results

In this section, the findings and analysis from empirical investigations are presented with regard to the three research questions in focus as stated in the introduction.

6.1. RQ (i) what are the main safety buffer sub-types utilized in manufacturing companies?

The empirically utilized safety buffer sub-types in the four manufacturing companies have been categorized according to the conceptual framework outlined above. The categorization is presented in Table 3, with companies that mention the safety buffer sub-type as described in the conceptual framework and an exemplifying quote from one of the respondents. Note that the purpose of each safety buffer sub-type for the companies comes forward in several quotes.

The selection of safety buffer sub-types varies between the companies. Some companies utilize almost all sub-types of safety buffers, while others, for different reasons, have a smaller set of safety buffer sub-types. Company C's respondents exemplified the difference in strategy for procurement depending on how critical the incoming material is and how close the supplier is in terms of geographical distance. For a closely located supplier, an inbound safety lead time buffer (B2) of 1.5 days is the only protection, while an inbound safety stock buffer (B1) covering eight weeks of demand is utilized for materials supplied across the world even though weekly deliveries are present. Wu and Holguin Veras (2022) found parts shortage as the second-most important factor for disruptions that effects delivery performance, which supports the strategy to utilize a larger safety for suppliers located overseas.

Three of the safety buffer sub-types were used by all companies: capacity flexibility buffer (B4), queue lead time buffer (B5), and outbound safety lead time buffer (B7). It is not a surprising result that all the companies utilize queue lead time buffers and capacity flexibility buffers, as it is common to consider queue lead time in capacity management (Vollmann et al., 2005), and various types of flexibility (e.g. overtime and

Table 3. Safety buffer sub-types utilized in practice.

#	Safety buffer sub-type	Used by	Supporting quote
B1	Inbound safety stock buffer	Company A, Company B, Company C	Company B, R5: 'The safety stock in procurement is individual … the level varies depending on the person, it can vary extensively.'
B2	Inbound safety lead time buffer	Company A, Company C, Company D	Company D, R11: 'To order [from suppliers] in advance is also a buffer … earlier ordering protects against the transport/delivery variations that are present.'
B3	Safety capacity buffer	Company B, Company D	Company D, R10: 'It [slack capacity] is a way to ensure that we get a reasonable lead time … '
B4	Capacity flexibility buffer	Company A, Company B, Company C, Company D	Company C, R8: 'We have labour exactly to reach the agreed service level agreement with sales … overtime is the option, it is our buffer in this.'
B5	Queue lead time buffer	Company A, Company B, Company C, Company D	Company A, R1: ' … have some form of lead time buffer, queue lead time for operations … with the main reason to protect the material flow [absorb variations].'
B6	Outbound safety stock buffer	Company B, Company D	Company D, R11: 'A safety stock that protects against variations in supply and demand.'
B7	Outbound safety lead time buffer	Company A, Company B, Company C, Company D	Company B, R5: ' … [lead time buffer] to absorb all problems … protects delivery with one day.'

temporary employment) are frequently utilized in practice for short-term requirements (Hedvall, Wikner, and Hilletofth 2017).

6.2. RQ (ii) what is the purpose of utilizing these safety buffer sub-types in manufacturing companies?

It is possible to extract several reoccurring nuances of purposes for the utilized safety buffer sub-types in practice. In Table 3, this takes the form by statements such as 'to protect the material flow', 'get a reasonable lead time', 'protects against the transport/delivery variations', 'absorb all problems', 'protects delivery' and 'protects against variations in supply and demand'. In general, protection appears to be the main purpose for utilizing safety buffers, which is rather self-evident. R5 argued that the overarching purpose, independent of the safety buffer type or sub-type, is to protect customer order deliveries. When asked further about this, R5 explained that the protection of customer order deliveries is the main concern, but if the materials, resource/work-centre capacity, and lead time are protected, the customer order deliveries will be protected as well (i.e. reduce a dimension of complexity by utilizing safety buffers, or additional resources as described by Tortorella et al. [2022]). Thus, the purpose can be defined in different ways depending on the point of view. See Table 4 for an overview of purposes as perceived in practice for the identified categories and safety buffer sub-types.

In general, inbound safety stock buffers before a resource/work centre (B1) are considered to enable high capacity utilization by protecting the resources from capacity losses, corresponding to a constraint buffer in the theory of constraints (TOC). The process buffers (B3, B4, and B5) are seen as protecting against capacity losses. Company D's respondents mentioned that the purpose of these safety buffer sub-types is to protect against stochastic variations in the manufacturing processes such as quality defects, capacity variations or demand variations. All lead-time safety buffers (B2, B5, and B7) are regarded as protecting subsequent processes, both processes in manufacturing by B2 (to absorb stochastic variations in delivery from suppliers) and the process of delivery by B5 and B7 (to absorb stochastic variations in the manufacturing processes). Both outbound buffers, B6 and B7, are considered safety buffer sub-types that compensate for capacity losses in the manufacturing processes to protect delivery. This is in line with the proposed purpose in the conceptual framework. Except for compensating capacity losses, R1 mentioned that the outbound safety lead time buffer (B7) is present due to practical planning purposes at the same time as it is a hedge against late deliveries that would lead to fines.

R1 argued that a safety buffer is something that is possible to use when needed, and it should be sustainable to use (i.e. the cost versus benefit). An important notion is that several issues affect the selection of safety buffer sub-types. R1 raised the issue of operations complexity, where it might not be possible to utilize external resources for increased capacity when there is a need for a certain resource or competence. R8 added that this issue remains even within Company C's manufacturing, where some resources/work centres might have extra capacity, but the personnel cannot contribute to other manufacturing processes due to limited physical abilities in relation to the heavy work required.

6.3. RQ (iii) what are the main concerns in the selection of safety buffers in manufacturing companies?

The total selection of safety buffer sub-types is an important aspect to consider in safety buffer management (Slack and Lewis, 2001). Company D's respondents emphasized the challenge of managing the safety buffers properly and maintaining them for what they are intended for, to handle stochastic variations, and not to utilize them for other challenges. This is especially common for lead time safety buffers, according to R11 who argued that B7 has been used by managers in various ways (e.g. change priorities between orders) because the planned date for finalizing an order is not corresponding to a late delivery to a customer. This mismatch between safety buffer purpose and utilization in practice leads to the first proposition:

Table 4. Safety buffer purposes in practice.

#	Category	Overarching purpose	Safety buffer sub-type	Purpose
X	Safety buffers	Protect the delivery capability	–	–
B1	Inbound buffers	Protect/enable high capacity utilization	Inbound safety stock buffer	Protection for quality defects, missing articles and late deliveries from affecting planned start of manufacturing.
B2			Inbound safety lead time buffer	Absorb late deliveries from affecting planned start of manufacturing.
B3	Process buffers	Protect against capacity losses/delays due to quality defects, capacity availability or stochastic demand variations	Safety capacity buffer	Additional capacity to absorb peak demand.
B4			Capacity flexibility buffer	Increase capacity availability when needed.
B5			Queue lead time buffer	Absorb stochastic variations that arise in the manufacturing processes.
B6	Outbound buffers	Compensate for capacity losses and protect delivery to customer	Outbound safety stock buffer	Secure high service level.
B7			Outbound safety lead time buffer	Absorb capacity losses from affecting delivery to customer.

1. P1: A lack of knowledge and clarity in safety buffer purposes increase the probability that the safety buffers are utilized for the wrong reasons.

Company Ás respondents emphasized that their safety buffer management, and safety buffer selection in particular, consists of active choices, but from different parties within the company (i.e. silo-thinking according to Fernández Campos, Huaccho Huatuco, and Trucco [2022]). According to R1, the internal efforts of mapping the existing safety buffer sub-types led to new insights of additional safety buffer sub-types utilized compared to what was known before. This situation seemed to be common across several cases, indicated by the number of safety buffer sub-types utilized in RQ (i). Company D's respondents added to the discussion by Company A's respondents and raised concerns of sub-optimization, where the selection of safety buffers is made locally without understanding the bigger picture. This is probably a result of each department working to secure its own performance measures without the holistic picture of what its decisions imply for the total. It could also be a result of complexity reduction, reducing the risk of delays as indicated by Tortorella et al. (2022). Hence, the total amount of safety buffers may be more than necessary and of the wrong type and sub-type from a holistic point of view, leading to Proposition 2 and Proposition 3:

2. P2: A lack of a holistic view in selecting safety buffers increases the probability of having too many and too large safety buffers.
3. P3: A lack of a holistic view in selecting safety buffers increases the probability of sub-optimised selections.

A lack of theoretical support for the selection of safety buffers is expressed as a common challenge across all cases. According to Krajewski et al. (2013), an intuitive approach is common when there is a lack of theoretical knowledge regarding how to evaluate the impact of different strategies. Company A has decided to have high levels of all their safety buffers in a transition period to takt-based production. R1 explained that the transition is a strategy to increase productivity and reduce lead times, where several safety buffer sub-types are built into the system to stabilize the material flow before starting to adjust the safety buffers. The lack of theoretical support for selecting safety buffers has been indicated together with a tendency to select and prefer a simple safety buffer sub-type and dimensioning method (i.e. method to determine the size of the safety buffer sub-type) within the sub-type over safety buffer sub-types and dimensioning methods of more advanced character. This leads to an additional proposition:

4. P4: A lack of support for selecting safety buffers increases the probability that intuitive approaches are utilized.

Slack and Lewis (2001) argued that the issue over time is determining when to make changes in the size of safety buffers, how big the changes should be and how soon changes should be performed. This study indicates that there are also challenges in deciding the overall selection of safety buffer sub-types, knowing the combinatorial effects of a certain selection of safety buffer sub-types, and when to use which safety buffer sub-type.

7. A conceptual framework for selecting safety buffers in practice

In total, seven different sub-types of safety buffers have been identified with different purposes. Based on logical reasoning, the frame of reference, and empirical results, a conceptual framework for selecting safety buffers in practice is presented in Table 5. The results are summarised as strengths and weaknesses with different safety buffer sub-types, with the objective to provide a rationale for the selection of safety buffer(s).

Nudurupati et al. (2022) emphasized the impact of performance measures on decision-making processes, thereby also the selection of safety buffers, meaning that certain safety buffers might be preferred based on used performance measures but could potentially inhibit the best choice from the strategic point of view. An important notion is that it is not always a question of choosing one safety buffer type or sub-type over the other, especially for safety buffers that can have the same type of effect on results. Instead, it can be choosing several safety buffers to different degrees, i.e. to balance them by the size of the safety buffers. In addition, Land et al. (2021) have demonstrated that safety buffers have an internal hierarchy and that they can be absorbed by other buffers or safety buffers. This can also be generalized to hierarchical decision-making where an adaptive approach can readily be employed to manage safety buffers (Wikner, Hedvall, and Mattsson 2019).

8. Implications for practice

According to Bish, Muriel, and Biller (2005), manufacturers have often used flexibility in resources to hedge against long-term forecast uncertainty while utilizing inventory to buffer against short-term variations. Managers are often evaluated on efficiency, resulting in high utilization rates leading to large amounts of inventory throughout the system as negative consequences (Cook 1994). Capacity cannot be stored in its native form, and management accounting systems thus report excess capacity as a cost and associate it with unfavourable performance (Fry, Steele, and Saladin 1994; Hristov et al. 2022) instead of a trade-off with lead time (Steele and Papke-Shields 1993). This inevitably leads to a tendency to favour material safety buffers over capacity safety buffers, while the capacity safety buffers and lead time safety buffers can be of more long-term strategic importance (Fry, Steele, and Saladin 1994). This is in line with findings by Rappold and Yoho (2008) that observed that senior management periodically shift focus to 'optimize' either customer service, capacity utilization or inventory levels instead of striving for a good balance. However, Hristov et al. (2022) found that unused capacity is seen as a risk that managers

Table 5. Framework of strengths and weaknesses with different safety buffer sub-types.

#	Safety buffer type	Safety buffer sub-type	Strengths	Weaknesses
B1	Material	Inbound safety stock buffer	+ Safety buffer in terms of materials making it preferable for stochastic variations in quantity. + Safety buffer that is immediately available for delivery. + Safety buffer that is preferable for low-cost articles.	− Results in higher holding costs compared to B2. − Safety buffer that is less suitable for high-cost articles with low order frequency.
B2	Lead time	Inbound safety lead time buffer	+ Safety buffer in terms of time making it preferable for stochastic variations in timing. + Safety buffer that is beneficial for lower-level articles and high-cost articles with low order frequency.	− Safety buffer that extends lead times compared to B1, resulting in longer delivery lead times and extends the frozen planning horizon. − Safety buffer that might create distorted priorities. − Safety buffer that is less suitable when delivery speed is important. − Articles are not immediately available for stochastic variations in quantity, making it less suitable for low-cost articles with high order frequency.
B3	Capacity	Safety capacity buffer	+ Safety buffer that reduce lead times, WIP and "waste", resulting in increased responsiveness. + Safety buffer with increased flexibility that can be used for a wider set of demands (compared to safety stock). + When the safety buffer is not used (for staff) there is a potential time to use for improvement work.	− Results in higher fixed costs compared to lead time safety buffers and material safety buffers. − Results in risk of resource loss/low capacity utilization when not used.
B4	Capacity	Capacity flexibility buffer	+ Safety buffer that results in increased responsiveness. + Safety buffer with increased flexibility that can be used for a wider set of demands (compared to safety stock). + Safety buffer that is preferable for stochastic variations in mix.	− Results in higher training costs compared to B3. − Results in higher costs for staff if overtime is used compared to B3. − Potentially results in higher fixed costs compared to B3 for equipment with high flexibility.
B5	Lead time	Queue lead time buffer	+ Safety buffer in terms of time making it preferable for stochastic variations in timing and in load.	− Safety buffer that extends lead times compared to B3 and B4, resulting in longer delivery lead times.
B6	Material	Outbound safety stock buffer	+ Safety buffer in terms of materials making it preferable for stochastic variations in quantity. + Safety buffer that is immediately available for delivery. + Safety buffer that is preferable for end-items dependent on high service level and reliability.	− Results in higher holding costs compared to B7. − Safety buffer that is less suitable for high-cost articles with low order frequency.
B7	Lead time	Outbound safety lead time buffer	+ Safety buffer in terms of time making it preferable for stochastic variations in timing. + Safety buffer resulting in high on-time delivery performance. + Safety buffer that is suitable when late deliveries lead to fines.	− Safety buffer that extends delivery lead times compared to B6 and extends the frozen planning horizon. − Safety buffer that might create distorted priorities. − Safety buffer that is less suitable when delivery speed is important. − Articles are not immediately available for stochastic variations in quantity, making it less suitable for low-cost articles with high order frequency.

are often measured on and that it affects the decision-making, but that it at the same time is still prioritized in terms of working climate and that it leads to better employee performance, justifying safety capacity as a safety buffer. In other words, an overview of possible safety buffer types, safety buffer sub-types and their purposes, strengths and weaknesses, which is provided in this research, can contribute to more rational decision-making and therefore improved safety buffer management. In our research project, the respondents emphasised that an increased insight into when different safety buffers are preferable was achieved. By clarifying the purposes, strengths and weaknesses of different safety buffers it can also facilitate the selection of safety buffers in practice. This recognition and awareness can also contribute to decision-making for increased responsiveness, which contributes to sustaining the future competitiveness of the business.

9. Theoretical contribution

The main theoretical contribution of this study is the investigation of safety buffer types and sub-types and their

purposes for manufacturing companies that resulted in four propositions, and a conceptual framework for the selection of safety buffers. No previous research has synthesized the different safety buffer types, sub-types and their characteristics into a framework, at least to the best of our knowledge, and is therefore a contribution to the safety buffer management literature. The intention is to increase the understanding that the selection of safety buffers is a critical capability. Slack and Lewis (2001) highlighted timing and magnitude of change in safety buffers as the main issues over time, while this research has also indicted challenges in deciding when to use which sub-type of safety buffer, the overall selection of safety buffers to use and implications of combinatorial effects of a certain selection of safety buffers.

The empirical results in this research have shown that capacity safety buffers have more nuances considered by practitioners in the safety buffer selection than expected, making the selection of safety buffers more challenging. One example is the impact of operational complexity for the capacity alternatives that are possible for the capacity safety buffer (e.g. a certain competence or capability required).

Employee involvement in improvement work has an indirect but significant impact on performance (Beraldin, Danese, and Romano 2022), which should be recognized to overcome the problems associated with the four propositions in this research (e.g. to increase awareness and reduce the risk of faulty utilization of safety buffers as in P1). Employees tend to self-organize resilience (Soliman and Saurin 2022), which leads to a tendency to have many safety buffers (P2) and the sub-optimization of these (P3). Ojiako et al. (2022) discussed ambiguity connected to performance management and the inconsistency and vagueness that can manifest related to what is measured and how, by whom, for what reasons, and the review frequency. Ambiguity in performance management is here considered a contributing factor for the occurrence of the propositions, especially P4 (i.e. a lack of support for selecting safety buffers leading to utilization of intuitive approaches). Ojiako et al. (2022) also found that it is not always clear why certain measures exists as the performance management systems emerge over time and that it is not always as purposefully designed and rational as assumed. This tendency seems to hold also for the selection of safety buffers, where both the utilized performance measures and human factors can lead to utilizing safety buffers for the wrong reasons (P1), having too many and too much safety buffers (P2), that the selections are suboptimized (P3) or not as systematic or rational as theoretically could be expected (P4).

10. Conclusions and further research

Stochastic variations can considerably degrade company performance (Rappold and Yoho 2008) and the selection of safety buffers is therefore of strategic importance. While the literature on selecting safety buffers is limited, related literature that addresses various aspects of safety buffer management is vast. It is beyond the scope of this study to cover all relevant literature in this area. However, this study describes a general research landscape and pinpoints possibilities of safety buffers for managing stochastic variations in manufacturing companies, thereby also addressing the call for more research that addresses how to cope with uncertainty and reducing the risk of delayed customer orders (Oehmen et al. 2009; Stefansson, Jensson, and Shah 2009). Oehmen et al. (2009) emphasized the need to identify, manage, and form strategies for apparent risks, where safety buffers play an important role for the risks associated with prevalent variations. In this research, a framework that conceptualizes important safety buffers has been established and illustrates the safety buffer types, their sub-types to consider, and their purposes. This is intended to contribute to an integrated perspective on safety buffers and with a more coherent terminology. Hence, a greater understanding of the contribution of safety buffers is achieved. In total, three different safety buffer types led to the identification of seven different safety buffer sub-types for manufacturing companies. Safety buffer management problems easily become complex, making superior solutions elusive (Boute et al. 2008). Combinatorial effects for utilizing different safety buffer types and sub-types, or many of the same type, should be considered to understand the relations better and the consequences of these relations as they affect the positioning and dimensioning of safety buffers.

The manufacturing companies in this study regard safety buffer management as an important challenge for the future competitiveness of their business. However, the respondents consider the decision support for safety buffer management to be insufficient. As a step towards developing decision support for safety buffer management, this study contributes to answering the questions of what, why, and how regarding selecting safety buffers. 'What?' refers to identifying the safety buffer sub-types utilized in manufacturing companies, 'Why?' refers to the purpose for each safety buffer sub-type, and 'How?' refers to the concerns for practical procedures with selecting safety buffers. The empirical findings support the existence of the various safety buffer sub-types in the developed conceptual framework. In conclusion of this study, four main concerns in safety buffer selection have been identified and expressed as four propositions and a conceptual framework was formed to facilitate the selection of safety buffers in practice by stating strengths and weaknesses with different safety buffer sub-types. Additional cases are needed to verify the generalisability of these propositions. These propositions also open potential areas for future research, e.g. why a holistic view is lacking for selecting safety buffers in manufacturing companies and how these concerns can be overcome.

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Appendix A *Workshop assignments*

Summary of assignment for workshop 1

The assignment is to identify safety buffers used in the selected unit of analysis (part of the company) based on the questions below. Three safety buffer types are described in the theoretical background to the assignment. Note that all these are not necessarily used in your selected unit of analysis. The answers to the questions should be presented during a 30-min presentation:

- Describe your selected unit of analysis based on the key components described in the assignment
- Identify the safety buffer types used in the selected unit of analysis. i.e. which safety buffer types are used in your selected unit of analysis? Why are these safety buffer types used?
- Analyse potential missing safety buffer type(s) in your unit of analysis. Do you miss any safety buffer type(s) in you unit of analysis? If yes, which one(s)? Are there any other safety buffer types than the three types mentioned in the assignment?
- Describe your experiences with the use of safety buffers. i.e. good and bad experiences from the use of safety buffers.

Summary of assignment for workshop 2

The assignment is to identify safety buffer types and sub-types used in the own company as a whole, with support from a conceptual framework of safety buffers. Note that the conceptual framework is just a baseline and should not be considered as complete. This means that not all safety buffer types or sub-types are necessarily used, and that additional safety buffer types or sub-types could be added for identified safety buffers in the company that do not correspond to the descriptions in the conceptual framework. In summary, the following questions should be answered during a 30-minute presentation:

- Which safety buffer types and sub-types are used? Identify the safety buffer types and sub-types that are used in the company with help from the conceptual framework.
- Why are these safety buffer types and sub-types used? Analyse each safety buffer type and sub-type that are used in regard to its purpose, and its effect(s) or consequence(s) for the company.

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